

Full solution packet for arXiv:1604.03831

The multiplier ideal $A_{(m)}^2$ is never maximal

Status. full_solution_likely_valid. This packet answers the maximal-ideal question in Section 5 of Andrzej Stanislaw Kucik, *Multipliers of Hilbert Spaces of Analytic Functions on the Complex Half-Plane*, arXiv:1604.03831.

Source question. Kucik constructs Hilbert spaces $A_{(m)}^2$ on the right half-plane by the Laplace transform

$$A_{(m)}^2 = \mathfrak{L}(L_{w_{(m)}}^2(0, \infty)), \quad w_{(m)}(t) = \sum_{n=0}^m 2\pi t^{2n} \int_0^\infty e^{-2rt} d\tilde{\nu}_n(r).$$

In Section 5, after assuming that $A_{(m)}^2$ is a Banach algebra, the source asks whether $A_{(m)}^2$ can be a maximal ideal in its multiplier algebra $\mathcal{M}(A_{(m)}^2)$. The source observes that maximality is equivalent to

$$\mathcal{M}(A_{(m)}^2) = A_{(m)}^2 + \mathbb{C}$$

and proves nonmaximality only in the special case $m = 1$ and $\nu_0 = \nu_1$.

For the remaining part of this paper we shall assume that $A_{(m)}^2$ is a Banach algebra.

Remark. If $A_{(m)}^2$ is an algebra, then it must, by definition, be an ideal in $\mathcal{M}(A_{(m)}^2)$. A natural question to ask here is: whether it could be a maximal ideal. If that was the case, then it would be the kernel of an algebra homomorphism, so $\text{codim } \mathcal{M}(A_{(m)}^2) A_{(m)}^2 = 1$, and since $A_{(m)}^2 \subset A_{(m)}^2 + \mathbb{C} \subseteq \mathcal{M}(A_{(m)}^2)$, we must have $\mathcal{M}(A_{(m)}^2) = A_{(m)}^2 + \mathbb{C}$. Conversely $A_{(m)}^2 + \mathbb{C}$ is the canonical unitisation of $A_{(m)}^2$, so if $\mathcal{M}(A_{(m)}^2) = A_{(m)}^2 + \mathbb{C}$, then $A_{(m)}^2$ must be a maximal ideal there. In the simplest case, when $m = 1$ and $\nu_0 = \nu_1$ we have that

$$\{g \in H^\infty(\mathbb{C}_+) : g' \in H^\infty(\mathbb{C}_+)\} \subseteq \mathcal{M}(A_{(1)}^2),$$

so $A_{(1)}^2$ cannot be a maximal ideal (take for example $e^{-z} \notin A_{(1)}^2 + \mathbb{C}$, which is obviously a multiplier). It still remains unclear whether there exists a sequence of measures $(\nu_n)_{n=0}^m$ such that $A_{(m)}^2$ would be a maximal ideal in the space of its multipliers.

Theorem 8. Suppose that for each $a > 0$ there exists $K > 0$ such that $w_{(m)}(t) \leq Ke^{at}$, for all $t > 0$. Let $\pi : \mathfrak{M}(A_{(m)}^2) \rightarrow \overline{\mathbb{D}}$ (the closed unit disk of the complex plane) be given by

$$\pi(\varphi) = \varphi\left(\frac{1-z}{1+z}\right) \quad (\varphi \in \mathfrak{M}(\mathcal{M}(A_{(m)}^2)))$$

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Nondegeneracy convention. The source defines $A^p(\mathbb{C}_+, (\nu_n)_{n=0}^m)$ with the norm obtained by summing the $A_{\nu_n}^p$ -norms of $F^{(n)}$, and calls it a Banach space. This source setup implicitly has $\tilde{\nu}_0 \neq 0$; otherwise nonzero constant functions would have norm zero in the ambient space. The theorem below is stated under this nondegenerate interpretation, which is also the setting of the source's standard examples.

Proof intuition. The missing obstruction is already present in the Laplace model. Multiplying a Laplace transform by e^{-az} translates the representing L^2 -function to the right by a . The zeroth measure gives a positive lower bound for the total weight $w_{(m)}$ near 0, and monotonicity of the individual Laplace transforms controls the weight ratio $w_{(m)}(t+a)/w_{(m)}(t)$ for large t . Therefore e^{-az} is always a multiplier. On the other hand, when $A_{(m)}^2$ is a Banach algebra, Kucik's necessary condition gives $A_{(m)}^2 \subset C_0(i\mathbb{R})$. The boundary function $e^{-ia y}$ does not converge at infinity, so e^{-az} cannot be an element of $A_{(m)}^2 + \mathbb{C}$. This prevents the multiplier algebra from being only the canonical unitization.

Lemma (Bounded horizontal shifts). *Assume $\tilde{\nu}_0 \neq 0$ and that the Laplace integrals defining $w_{(m)}$ are finite for every positive argument. Then for every $a > 0$ there is a constant $C_a < \infty$ such that*

$$w_{(m)}(t+a) \leq C_a w_{(m)}(t) \quad (t > 0).$$

Proof. Write

$$L_n(t) = \int_0^\infty e^{-2rt} d\tilde{\nu}_n(r), \quad w_n(t) = 2\pi t^{2n} L_n(t).$$

Each L_n is decreasing on $(0, \infty)$. Since $\tilde{\nu}_0 \neq 0$, $L_0(1) > 0$. Hence, for $0 < t \leq 1$,

$$w_{(m)}(t) \geq w_0(t) = 2\pi L_0(t) \geq 2\pi L_0(1) > 0.$$

Also, for $0 < t \leq 1$,

$$w_n(t+a) \leq 2\pi(a+1)^{2n} L_n(a),$$

so $\sup_{0 < t \leq 1} w_{(m)}(t+a)/w_{(m)}(t) < \infty$.

For $t \geq 1$, we have $t+a \leq (1+a)t$, and therefore

$$w_n(t+a) = 2\pi(t+a)^{2n} L_n(t+a) \leq (1+a)^{2m} 2\pi t^{2n} L_n(t) = (1+a)^{2m} w_n(t).$$

Summing over n gives the same bound for $w_{(m)}$ on $[1, \infty)$. The two estimates together prove the lemma. $\square$

Theorem. *Let $A_{(m)}^2$ be one of the source paper's nondegenerate spaces, with $\tilde{\nu}_0 \neq 0$. If $A_{(m)}^2$ is a Banach algebra, then it is not a maximal ideal in $\mathcal{M}(A_{(m)}^2)$. More precisely, for every $a > 0$,*

$$e^{-az} \in \mathcal{M}(A_{(m)}^2) \setminus (A_{(m)}^2 + \mathbb{C}).$$

Proof. Fix $a > 0$ and let $h_a(z) = e^{-az}$. Take $F = \mathfrak{L}f \in A_{(m)}^2$. Then

$$h_a(z)F(z) = e^{-az} \int_0^\infty f(s)e^{-sz} ds = \int_a^\infty f(t-a)e^{-tz} dt = \mathfrak{L}(S_a f)(z),$$

where $S_a f(t) = 0$ for $0 < t < a$ and $S_a f(t) = f(t-a)$ for $t \geq a$. By the lemma,

$$\|S_a f\|_{L_{w_{(m)}}^2}^2 = \int_0^\infty |f(s)|^2 w_{(m)}(s+a) ds \leq C_a \int_0^\infty |f(s)|^2 w_{(m)}(s) ds.$$

Thus multiplication by h_a is bounded on $A_{(m)}^2$, so $h_a \in \mathcal{M}(A_{(m)}^2)$.

It remains to show that $h_a \notin A_{(m)}^2 + \mathbb{C}$. Since $A_{(m)}^2$ is a Banach algebra, Kucik's Theorem 5 gives

$$L_{w_{(m)}}^2(0, \infty) \subset L^1(0, \infty) \quad \text{and} \quad A_{(m)}^2 \subset C_0(i\mathbb{R}).$$

Equivalently, each $F \in A_{(m)}^2$ has boundary values satisfying $F(iy) \rightarrow 0$ as $|y| \rightarrow \infty$. If $h_a = F + c$ with $F \in A_{(m)}^2$ and $c \in \mathbb{C}$, then

$$e^{-ia y} = F(iy) + c \quad (y \in \mathbb{R}).$$

The right-hand side has limit c as $|y| \rightarrow \infty$. The left-hand side has no limit: along $y = 2\pi k/a$ it is 1, while along $y = (2k+1)\pi/a$ it is -1 . This contradiction proves $h_a \notin A_{(m)}^2 + \mathbb{C}$.

The source remark shows that, under the Banach-algebra assumption, maximality of $A_{(m)}^2$ in its multiplier algebra would force $\mathcal{M}(A_{(m)}^2) = A_{(m)}^2 + \mathbb{C}$. Since h_a is a multiplier outside that unitization, $A_{(m)}^2$ is not maximal. $\square$

Novelty and literature check. The local run indexes were searched for the arXiv id, source title keywords, `multiplier`, `maximal ideal`, `A^2_(m)`, and `Corona`; no exact existing packet for this maximal-ideal question was found. A bounded external search on 2026-07-19 for 1604.03831 `maximal ideal`, the source title with `corona`, `A^2_(m) maximal ideal multipliers`, and Kucik/corona keywords found the journal version in *Operators and Matrices*, the author's 2017 thesis, and adjacent Laplace-Carleson papers, but no later exact answer to this question.

Limitations. The proof is a negative answer in the nondegenerate source setting. If a reviewer intentionally allows the degenerate case $\tilde{\nu}_0 = 0$, the argument no longer gives boundedness of every point-mass shift near the origin; that degenerate variant is not needed for the source's Banach-space setup but should be kept separate if studied.

Human-review recommendation. Review as a short full solution of the source's maximal-ideal question. The two points worth checking are the implicit nondegeneracy of $\tilde{\nu}_0$ in the source definition, and the elementary weight-ratio estimate in the shift lemma.

References

- [1] A. S. Kucik, *Multipliers of Hilbert Spaces of Analytic Functions on the Complex Half-Plane*, arXiv:1604.03831; published version: *Operators and Matrices* **11** (2017), no. 2, 435–453.